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On

EMOGALAXY

FACE RECOMMENDATION PLATFORM

COMPUTER SCIENCE AND ENGINEERING

B.E. Batch-2022

in

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ABSTRACT

EmoGalaxy is an advanced real-time emotion detection and analysis system designed to interpret human emotions using multimodal inputs such as facial expressions, voice signals, and behavioral patterns. The system leverages computer vision and machine learning techniques to analyze emotions through webcam feeds, uploaded images, and video inputs, providing accurate and dynamic emotion recognition.

The application integrates facial emotion detection using deep learning models, along with voice-based emotion analysis through acoustic feature extraction. A unique multimodal fusion mechanism combines these inputs to generate a more reliable and context-aware emotional output. Additionally, the system incorporates behavioral analysis by tracking emotional variations over time, identifying stability, volatility, and stress indicators in user behavior.

EmoGalaxy further enhances its capabilities with multi-person face tracking, enabling simultaneous emotion analysis for multiple individuals, and an emotion prediction module that forecasts future emotional states using probabilistic modeling techniques. The platform also provides interactive visualizations, detailed reports, and real-time analytics to improve user understanding and decision-making.

Designed with scalability, usability, and innovation in mind, EmoGalaxy serves as a comprehensive solution for emotion-aware applications in domains such as mental health monitoring, human-computer interaction, surveillance, and user experience research.

INDEX

1. INTRODUCTION

1.1 Background

In recent years, there has been a significant rise in the development of intelligent systems capable of understanding human behavior and emotions. With the advancement of artificial intelligence, machine learning, and computer vision technologies, systems are no longer limited to processing structured data but are increasingly capable of interpreting complex human signals such as facial expressions, speech patterns, and behavioral changes.

Emotion recognition has become a critical area of research due to its wide range of applications in domains such as mental health monitoring, human-computer interaction, surveillance, customer experience analysis, and virtual assistants. Traditional methods of emotion detection relied primarily on manual observation or single-modal systems, which often resulted in limited accuracy and lack of contextual understanding.

However, human emotions are inherently multimodal, meaning they are expressed through a combination of facial cues, vocal tone, and behavioral patterns. Relying on a single source of data, such as facial expressions alone, may lead to inaccurate or incomplete interpretations. This has led to the emergence of multimodal emotion recognition systems that combine multiple inputs to improve reliability and performance.

With the increasing demand for real-time and intelligent emotion-aware applications, there is a need for systems that can efficiently process multiple data sources, provide continuous analysis, and generate meaningful insights. EmoGalaxy is developed as a response to this need, integrating facial, voice, and behavioral analysis into a unified platform to deliver accurate and context-aware emotion detection.

1.2 Problem Statement

Despite rapid advancements in artificial intelligence and emotion recognition technologies, existing systems still face several limitations in accurately interpreting human emotions. Many traditional emotion detection systems rely on a single modality, such as facial expressions or text-based inputs, which often leads to incomplete or misleading results due to the complex and dynamic nature of human emotions.:

- **Limited Accuracy**: Systems based solely on facial expressions may fail in cases of low lighting, occlusion, or subtle emotional changes.
- **Lack of Multimodal Integration**: Most existing solutions do not effectively combine facial, voice, and behavioral data, resulting in reduced reliability.
- **Absence of Real-Time Analysis**: Many systems are not capable of providing continuous, real-time emotion tracking and feedback.
- **Inability to Handle Multiple Users**: Existing applications often fail to detect and analyze emotions of multiple individuals simultaneously.
    - **Lack of Predictive Capability**: Most systems only detect current emotions and do not predict future emotional states.
    - **Poor User Interaction and Visualization**: Limited visualization tools reduce the interpretability of emotional data.

2. Software and Hardware Requirement Specification

2.1 Methods

EmoGalaxy was developed following modern software engineering and artificial intelligence practices to ensure scalability, accuracy, and real-time performance:

- **Agile Development Methodology**: Iterative development approach was used to continuously improve features such as emotion detection, fusion, and prediction modules.

- **Modular Architecture**: The system is divided into independent modules such as facial emotion detection, voice analysis, emotion fusion, behavioral analysis, and prediction, ensuring maintainability and scalability.

- **AI-Driven Design**: Machine learning and deep learning techniques are used for emotion detection through facial expressions and voice signals.
- **Real-Time Processing Pipeline**: Streamlit and WebRTC are used to process live webcam and audio streams efficiently with minimal latency.
- **Multimodal Fusion Strategy**: Combines facial and voice emotion predictions using weighted averaging to improve accuracy and reliability.
- **Data-Driven Analysis**: Emotion data is processed and analyzed to generate insights such as volatility, stability, and stress indicators.
- **Visualization-Centric Design**: Interactive charts and reports using Plotly are integrated to enhance user understanding of emotional patterns

2.2 Programming/Working Environment

- Frontend / Interface Development
  - Framework: Streamlit
  - Streamlit-WebRTC
    - Styling: Custom CSS within Streamlit
    - Visualization: **Plotly, Matplotlib**
    - Development Tools: Visual Studio Code, Git
- Backend / Processing
  - Programming Language: **Python 3.11**
  - Computer Vision: **OpenCV**
  - Facial Emotion Analysis: **DeepFace** [1]
  - Audio Processing: **librosa, sounddevice**
  - Data Processing: **NumPy, Pandas**
  - Scientific Computing: **SciPy**
  - Machine Learning Support: **scikit-learn, TensorFlow/Keras[3]soun**
- Version Control and Deployment:

- Version Control: Git and GitHub
- Execution Platform: Local system / Streamlit server

2.3 Requirements to Run the Application

- Software Requirements:
  - Operating System: Windows / macOS / Linux
  - Python Version: Python 3.11 or above
  - Browser: Chrome, Firefox, Edge, or Safari
  - Required Libraries:
    - streamlit
    - opencv-python
    - deepface[1]
    - librosa
    - numpy, pandas
    - plotly, matplotlib
    - scipy, scikit-learn

- Hardware Requirements:
  - Processor: Intel i5 or higher (or equivalent)
  - RAM: Minimum 8 GB (16 GB recommended for smooth processing)
  - Storage: At least 1–2 GB free disk space
  - Camera: Webcam (for real-time emotion detection)
  - Microphone: Required for voice emotion analysis

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3. Database Analysis, Design, and Implementation

3.1 Database Overview

The EmoGalaxy system is designed as a multimodal emotion analysis platform that processes facial, vocal, and behavioral data to detect and interpret human emotions in real time. Unlike traditional systems that rely on a single input source, EmoGalaxy integrates multiple modalities to enhance accuracy and contextual understanding.

The system follows a pipeline-based architecture where input data (video/audio) is processed through multiple AI modules, and the results are combined to generate final emotional insights and predictions.

3.2 Database Design

The core architecture includes the following primary entities:

- **Face Emotion Detection Module**
  Uses DeepFace[1] and OpenCV to detect faces and classify emotions from facial expressions.

- **Voice Emotion Detection Module**
  Extracts audio features such as MFCC, pitch, and energy using librosa to determine emotions from speech.

- **Emotion Fusion Module**
  Combines face and voice predictions using weighted averaging to generate a final emotion output.

- **Behavioral Analysis Module**
  Analyzes emotion changes over time to calculate metrics like stability score, volatility index, and stress indicators.

- **Multi-Person Tracking Module**
  Tracks multiple individuals using centroid-based tracking and assigns unique IDs for separate emotion analysis.

- **Emotion Prediction Module**

  Uses a Markov Chain model to predict future emotional states based on historical emotion sequences

3.3 Key Relationships

- Facial and voice emotion modules independently generate emotion predictions.
- The fusion module combines both outputs to produce a final emotion.
- Behavioral analysis uses time-series emotion data for insights.
- Multi-person tracking associates emotions with specific individuals.
- Prediction module uses past emotional data to forecast future states

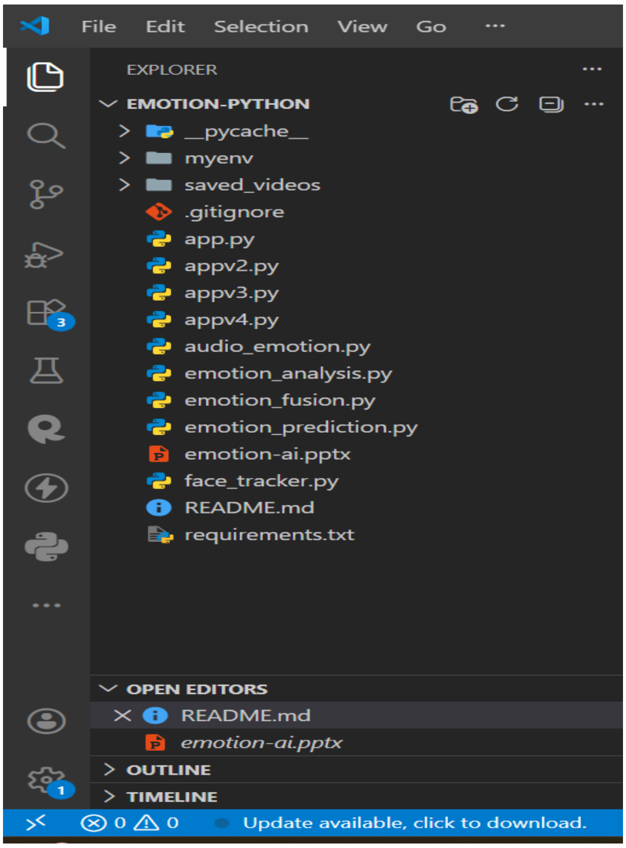
3.4 Data Relationships Summary

The system workflow can be summarized as follows:

- Input is received via webcam, image, or video upload.
- Face detection and voice processing are performed simultaneously.
- Emotion predictions are generated from both modalities.
- Fusion module combines predictions for the final output.
- Behavioral analysis and prediction modules process historical data.
- Results are displayed through interactive dashboards and reports.

4. Program's Structure Analysis and GUI Construction

4.1 Project Directory Structure:

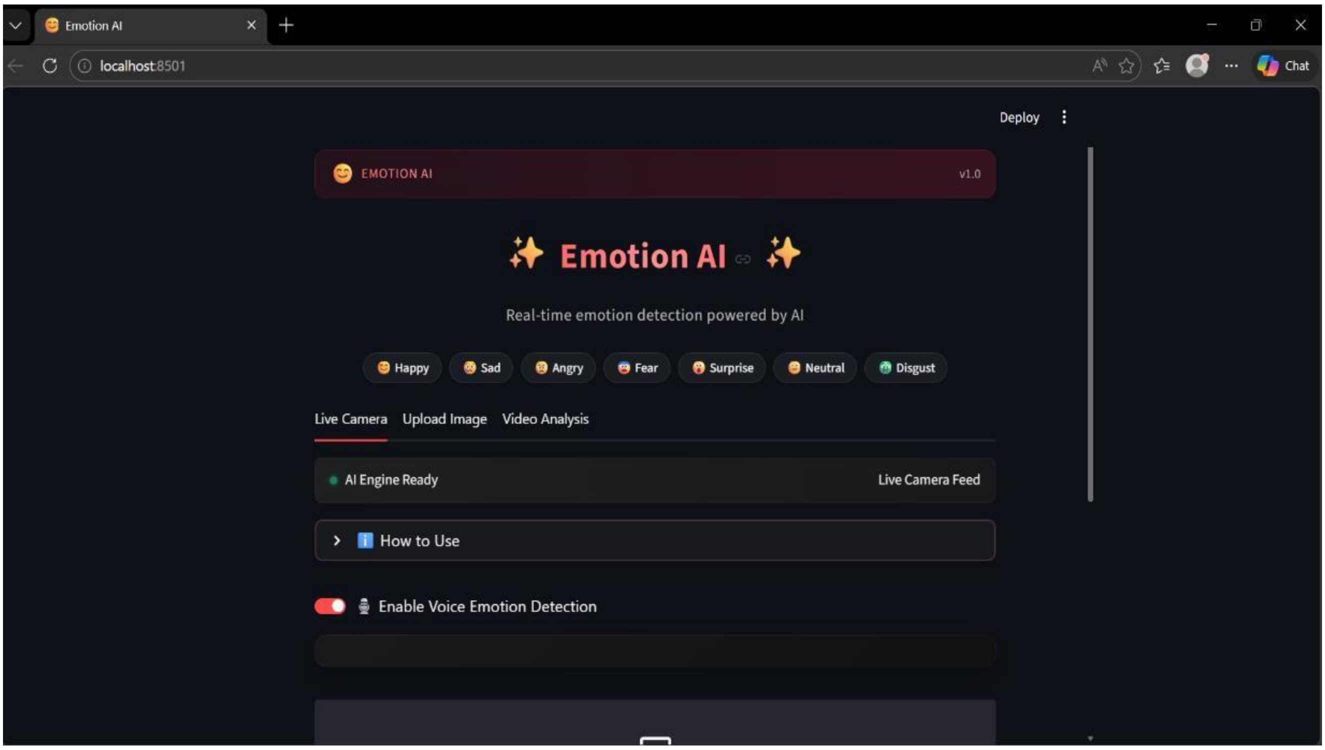


4.2 GUI CONSTRUCTING

The Graphical User Interface (GUI) of the Emotion AI system is built using Streamlit, offering a modern, interactive, and responsive design that enhances usability and engagement. Advanced styling with custom CSS and animated elements creates a sleek and professional user experience. The interface supports three main functionalities—live webcam detection, image upload, and video analysis, organized via tabbed navigation.

1. Home Interface

The home interface presents a clean and visually engaging layout with the application title EmoGalaxy”, along with intuitive navigation across different modules. It provides a structured entry point to access all functionalities and highlights the system’s capabilities in real-time emotion detection, analysis, and prediction.

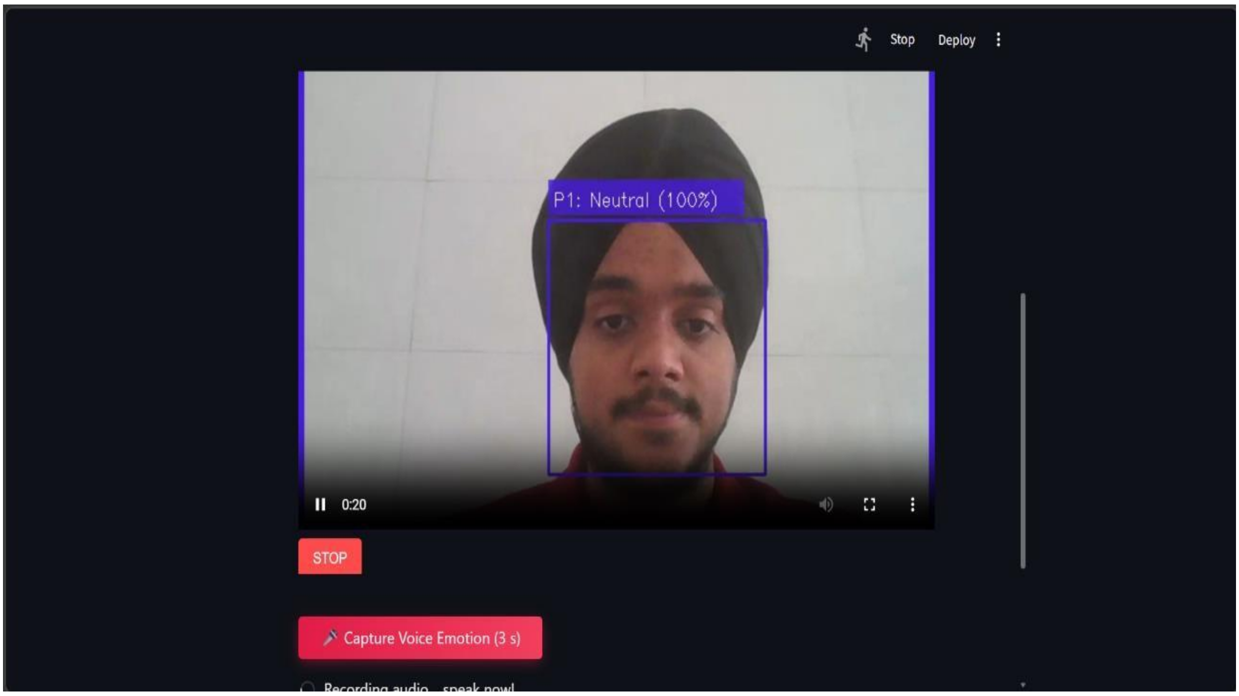


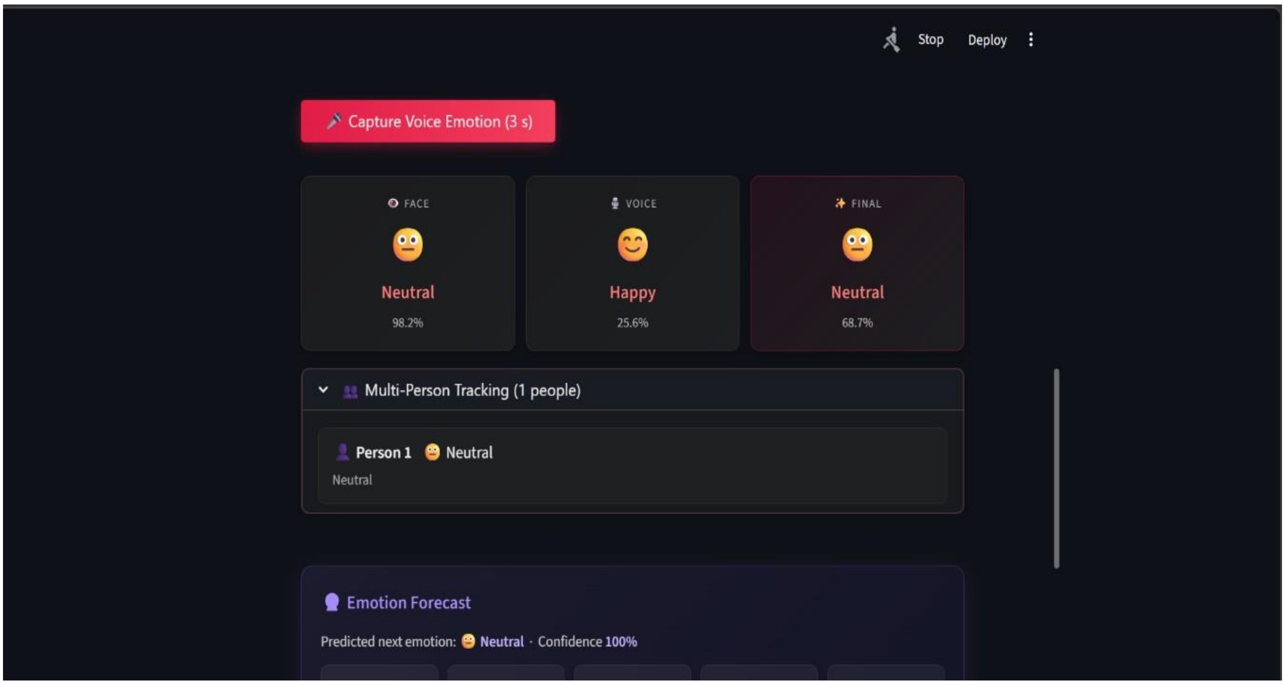
2. Live Camera Tab

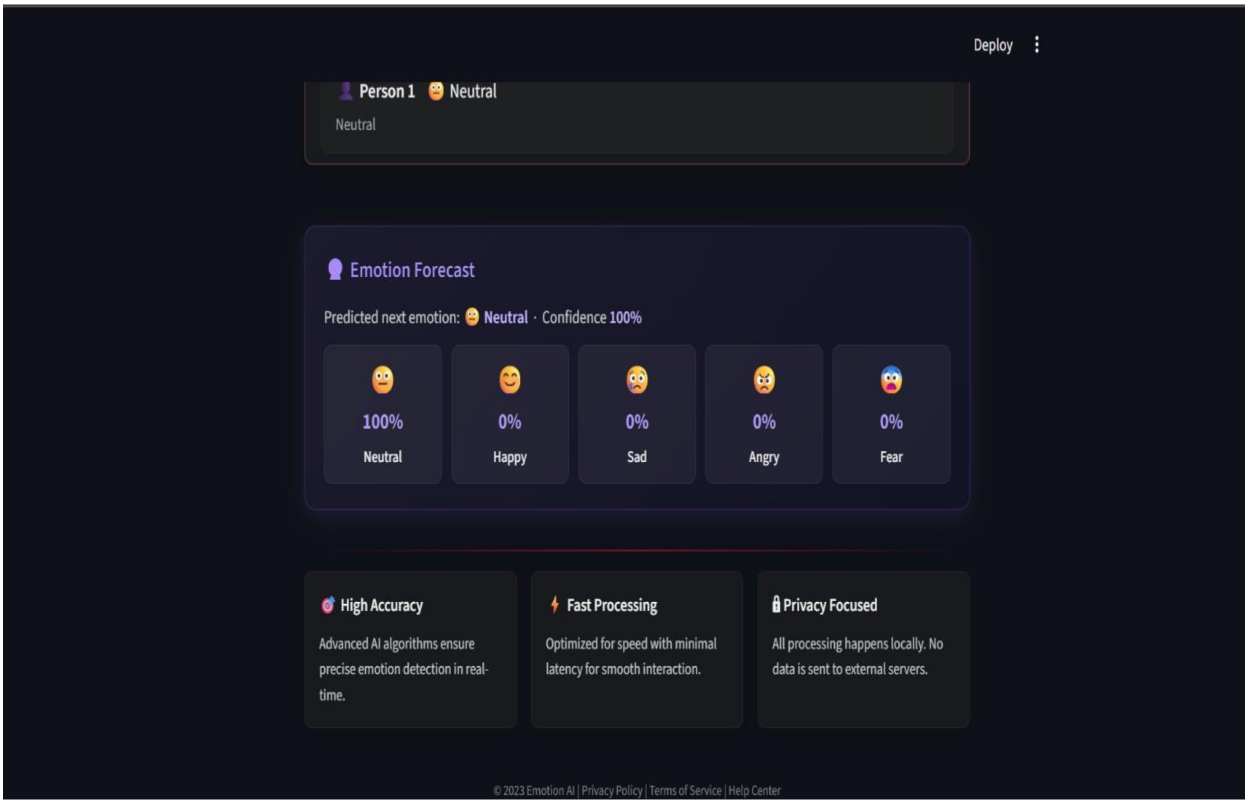
Enables real-time emotion detection using webcam and microphone input. This section includes:

- Live webcam feed with detected faces highlighted using bounding boxes
- Emotion labels displayed with confidence percentages
- Voice emotion capture option using microphone input
- Multimodal emotion display showing:
    - Face Emotion
    - Voice Emotion
    - Final Fused Emotion
- Real-time tracking of multiple individuals with unique IDs (e.g., P1, P2)
- Continuous emotion updates with smooth temporal transitions

- Interactive panels for monitoring live emotional states



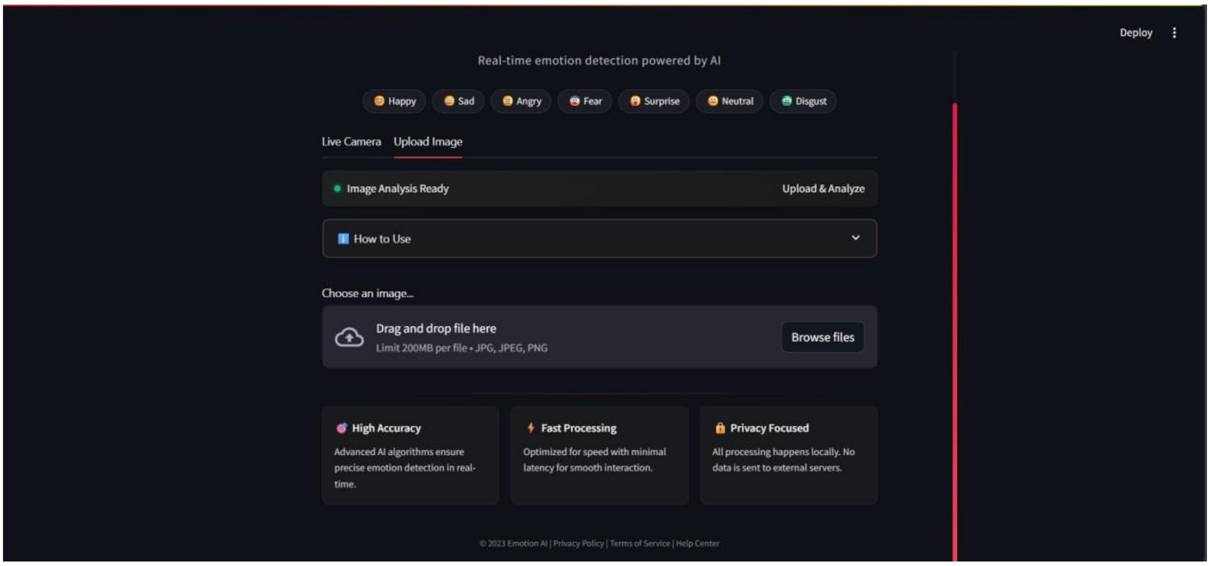


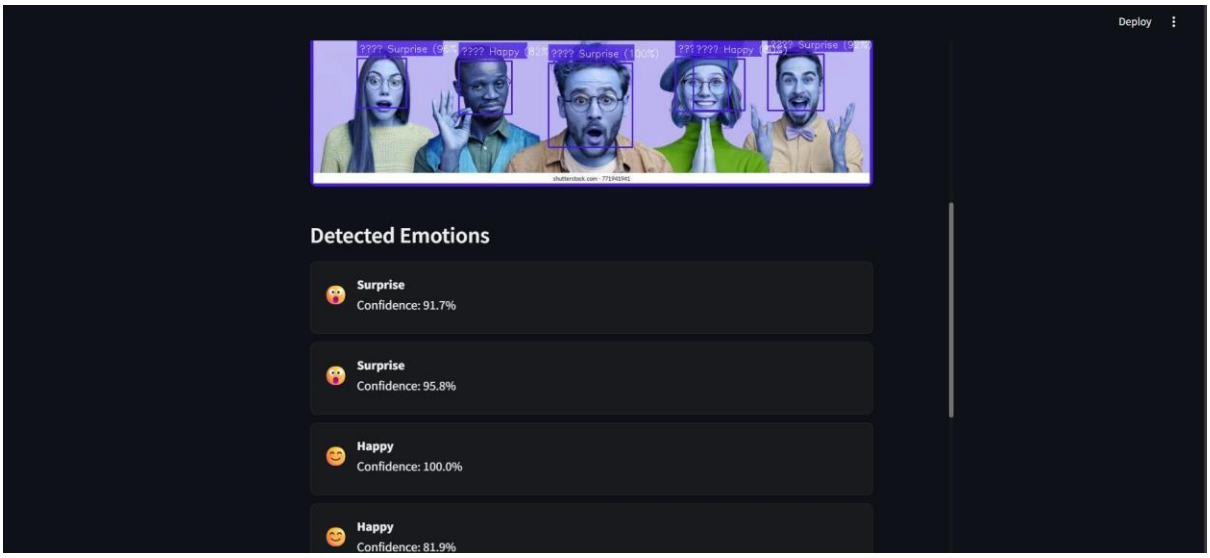


3. Upload Image Tab

Allows users to upload images for one-time emotion detection. Features include:

- Image preview before processing
- Detection of one or multiple faces in the image
- Emotion labels displayed for each detected face
- Confidence score visualization for detected emotions
- Clean and structured display of results for better readability





4. Video Analysis Tab

Provides advanced emotion analysis for uploaded video files. This section includes:

- Upload functionality for video formats (MP4, AVI, MOV)
- Adjustable sampling rate for performance optimization
- Detailed summary including:
  - Video duration
  - Number of frames analyzed
  - Dominant emotion
- Interactive visualizations such as:

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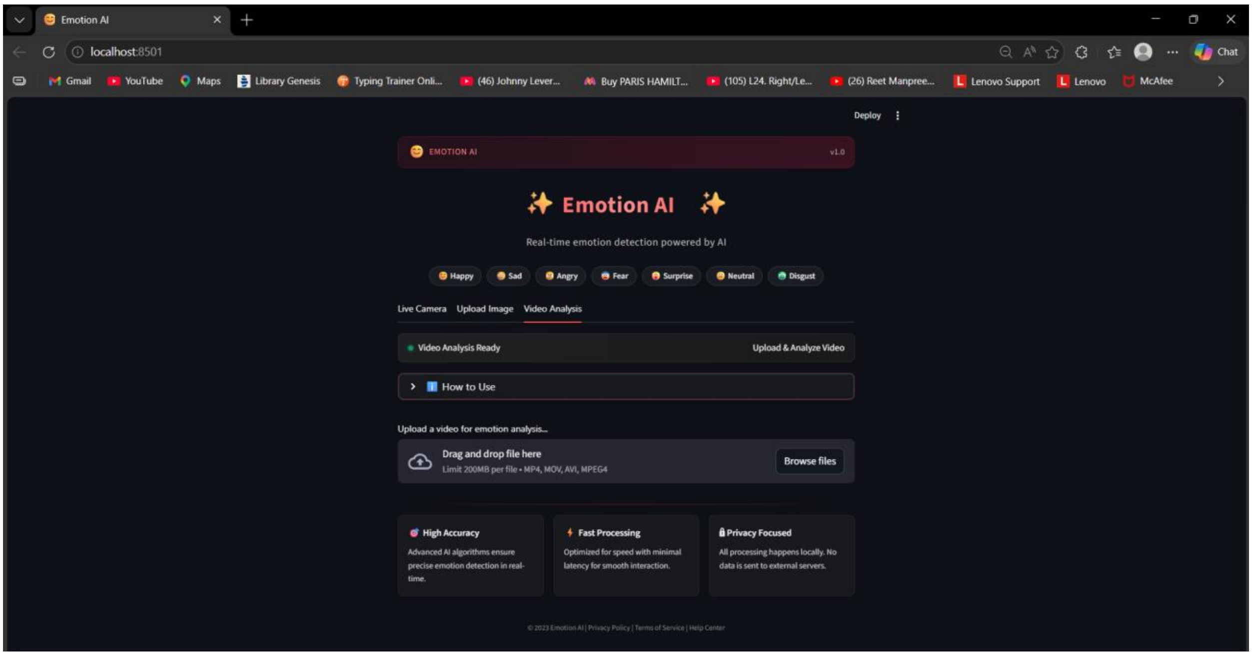
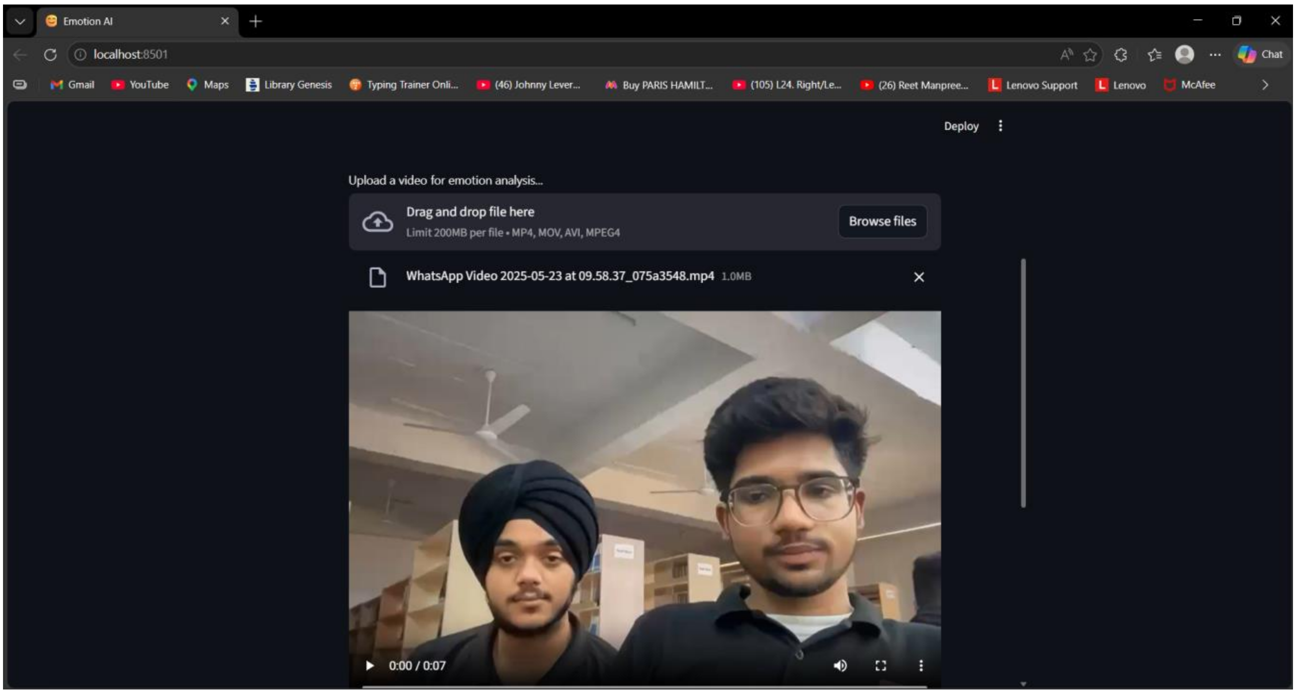
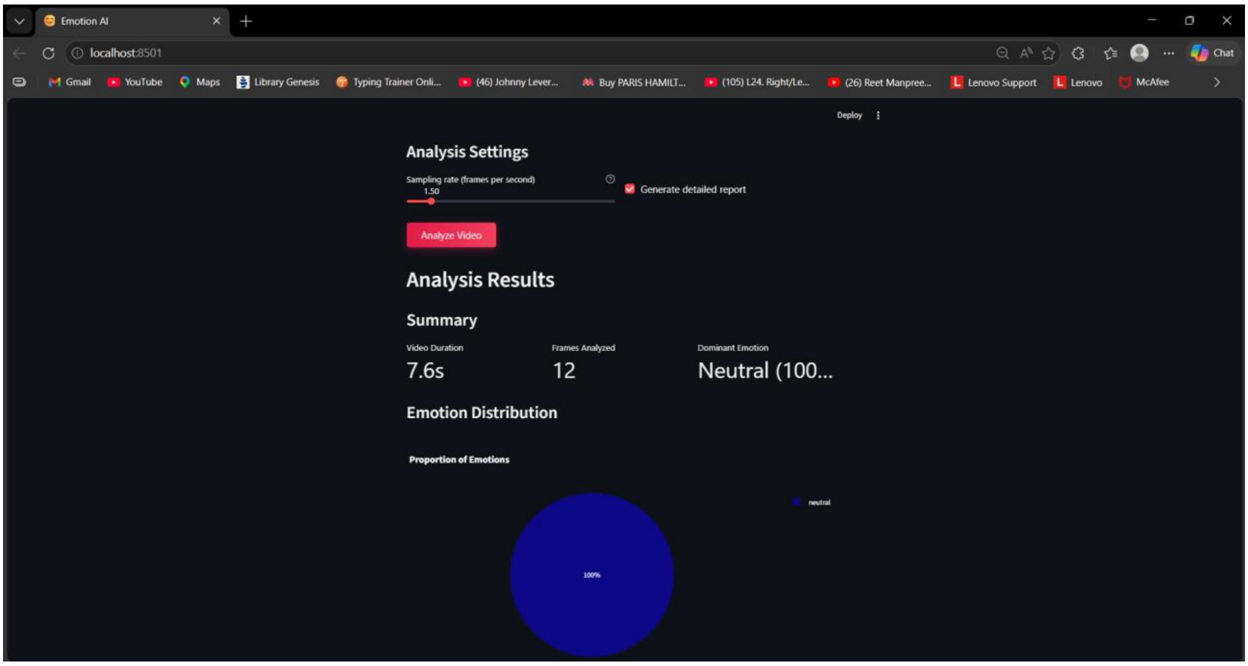
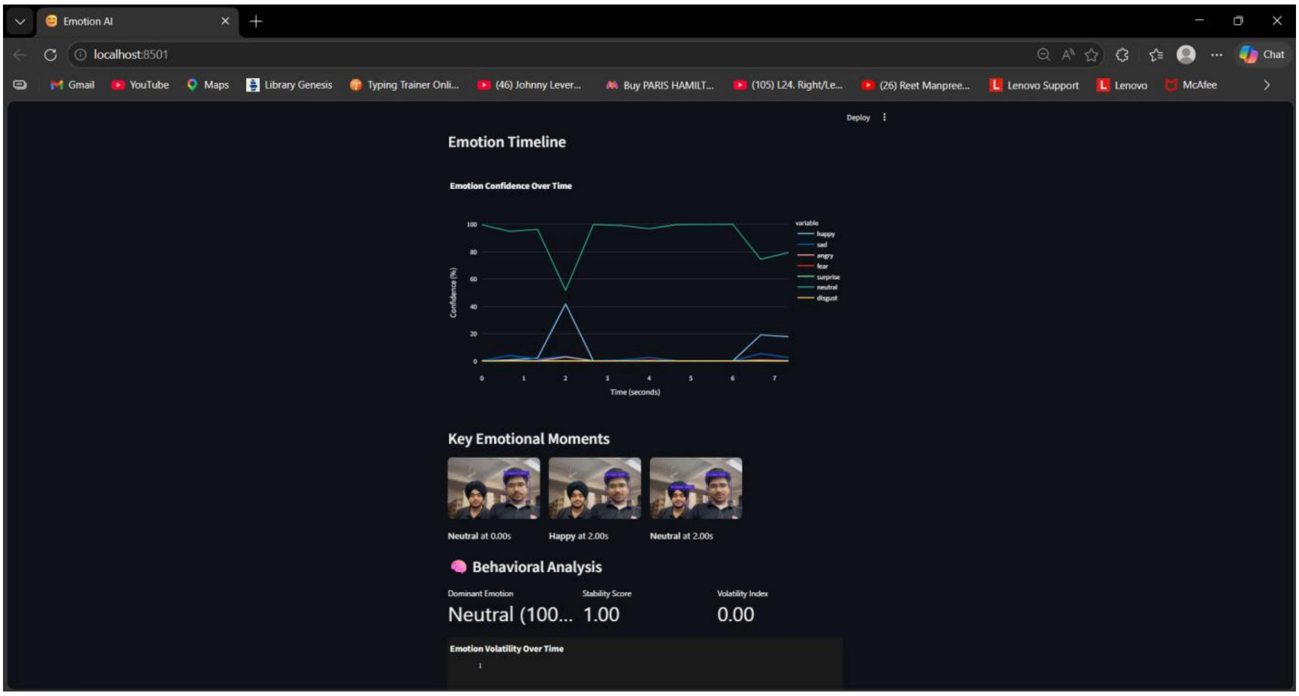
- Emotion distribution (pie chart)
- Emotion timeline (line graph)

- Detection of key emotional moments across frames







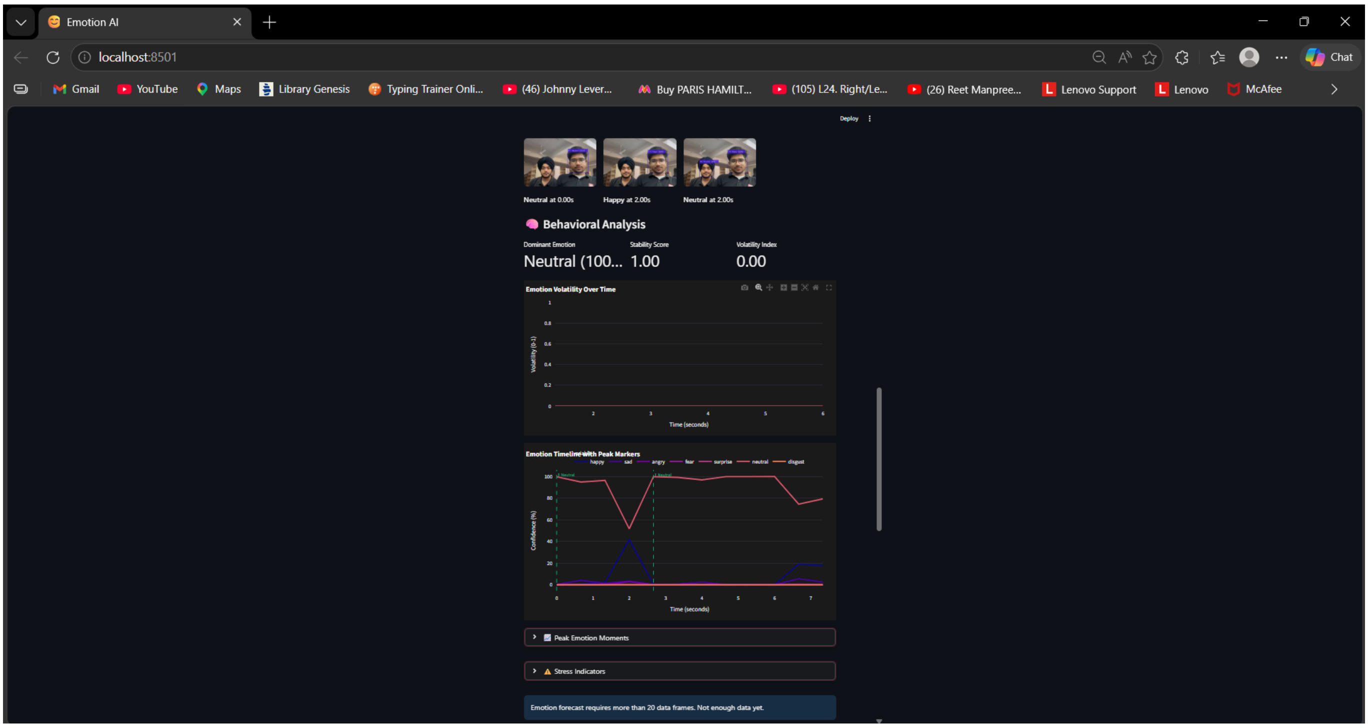


5. Behavioral Analysis Dashboard

Enhances video analysis by providing deeper insights into emotional patterns:

- Volatility Index indicating frequency of emotion changes
- Stability Score representing emotional consistency
- Detection of peak emotional moments (high-confidence emotions)

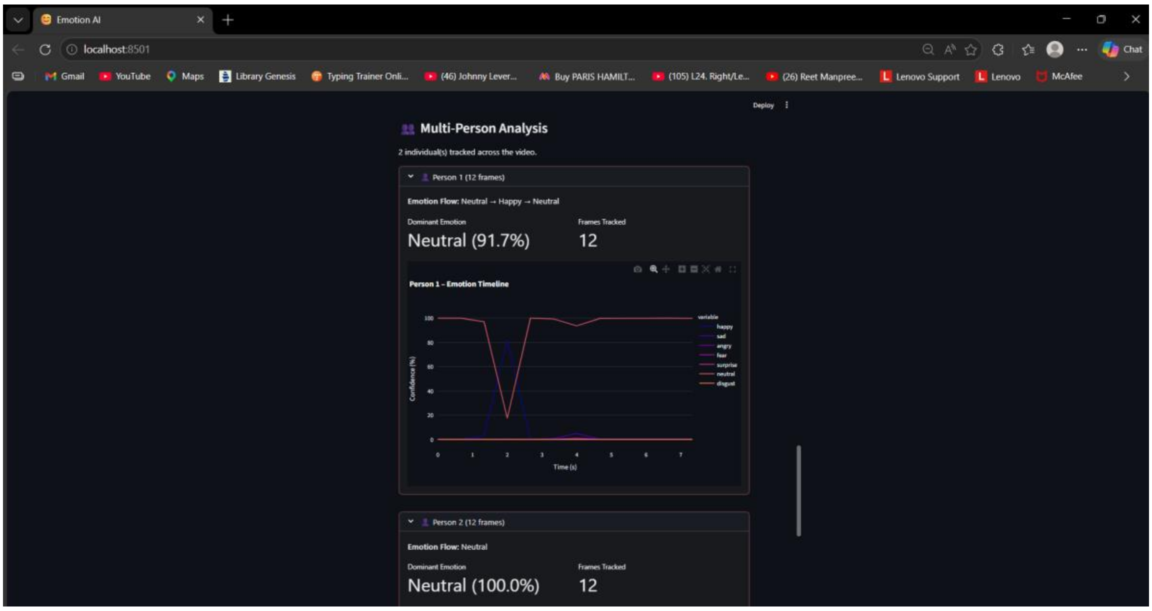
- Identification of **stress indicators** based on rapid emotion transitions
- Graphical representation of emotional trends over time

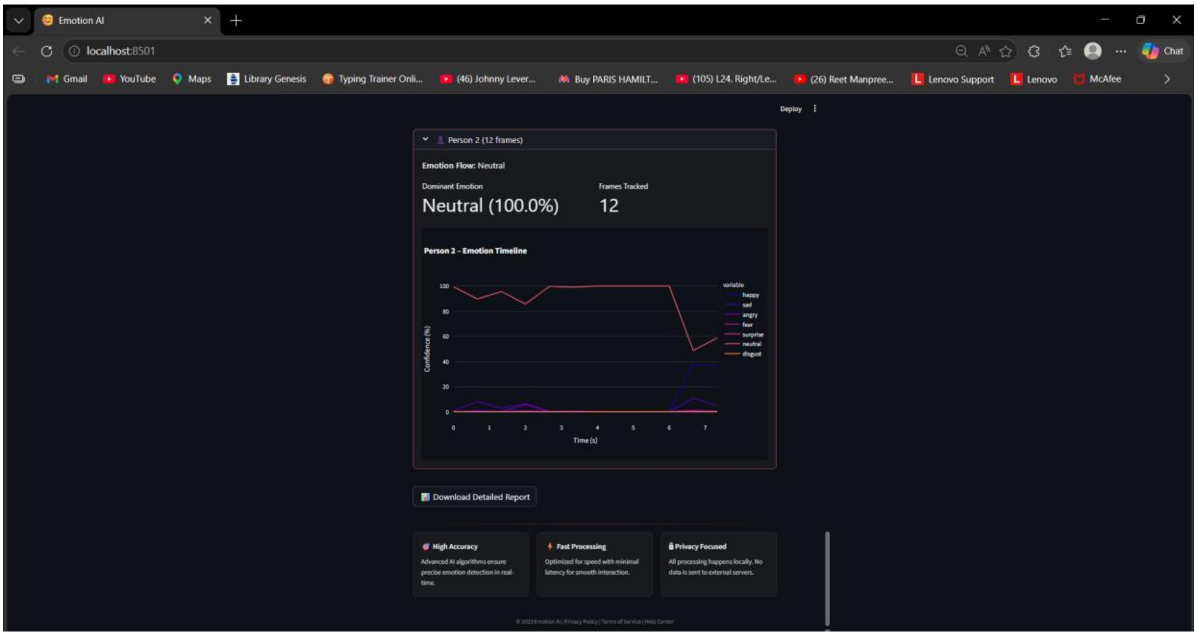


6. Multi-Person Analysis Interface

Supports simultaneous tracking and analysis of multiple individuals:

- Assigns unique IDs to each detected person
- Displays per-person emotion timeline and statistics
- Shows dominant emotion for each individual
- Visual separation of emotional data for clarity

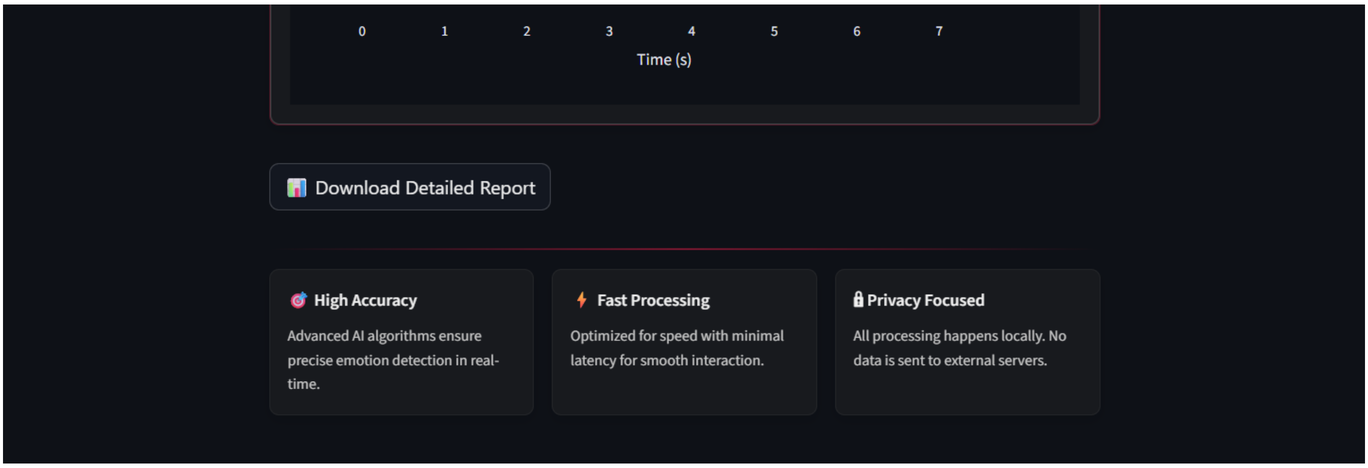


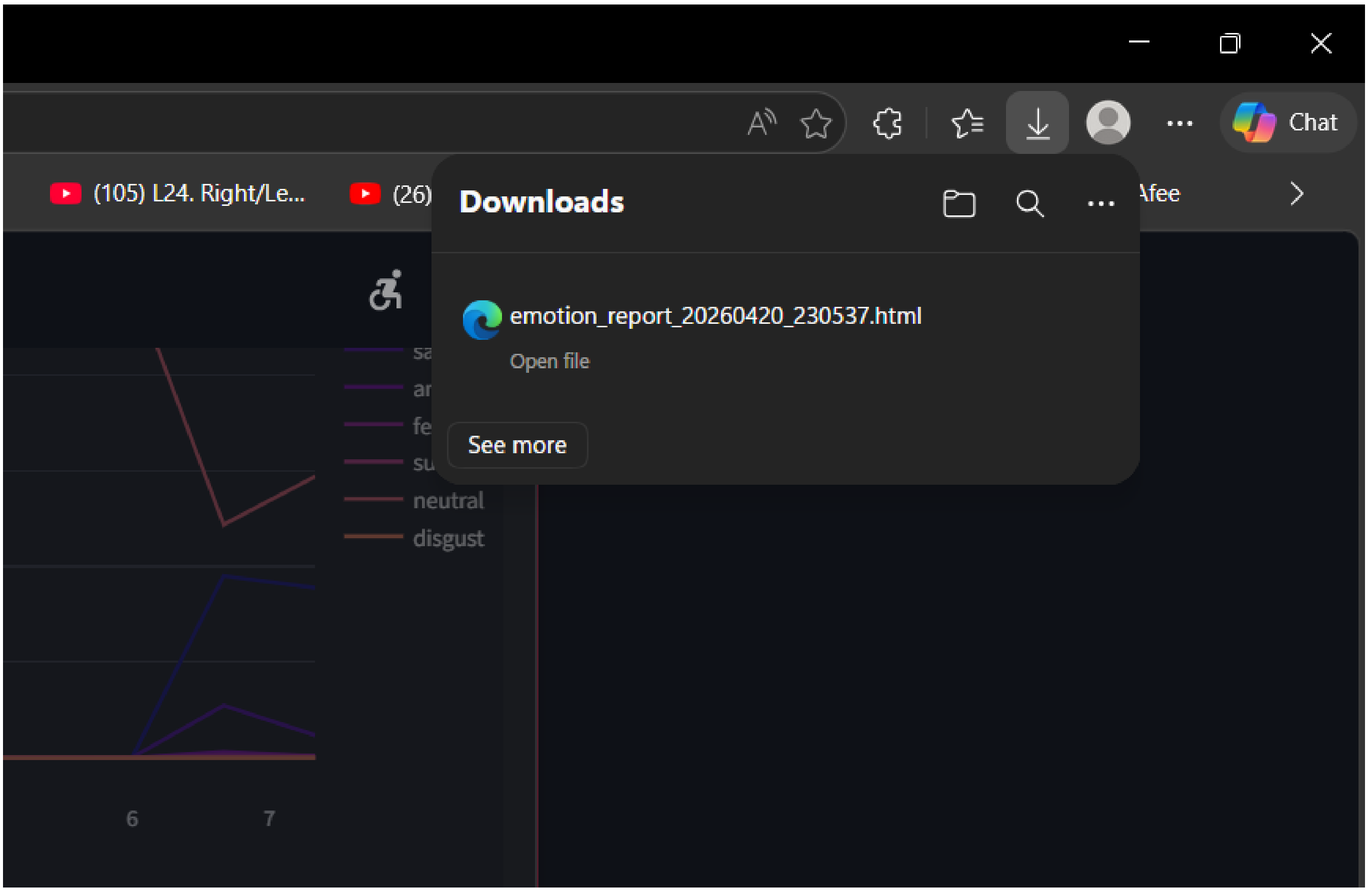


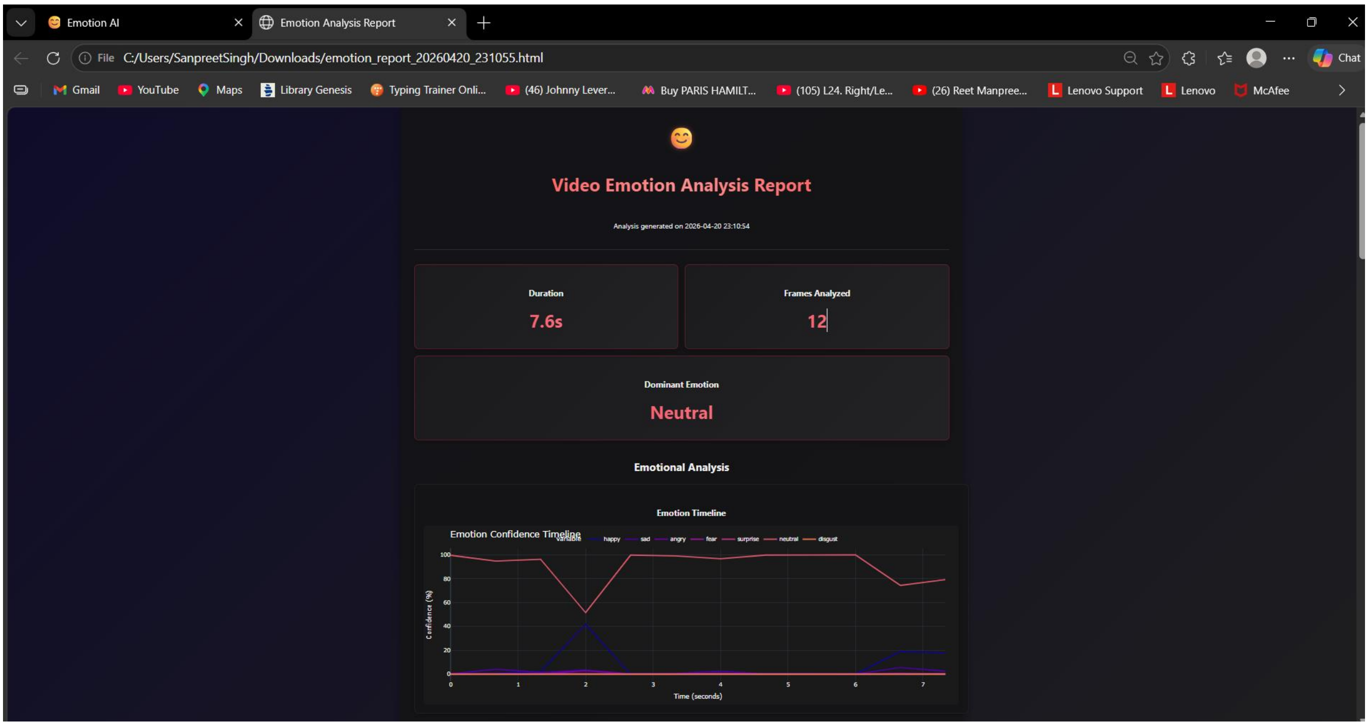
7. Report Generation Interface

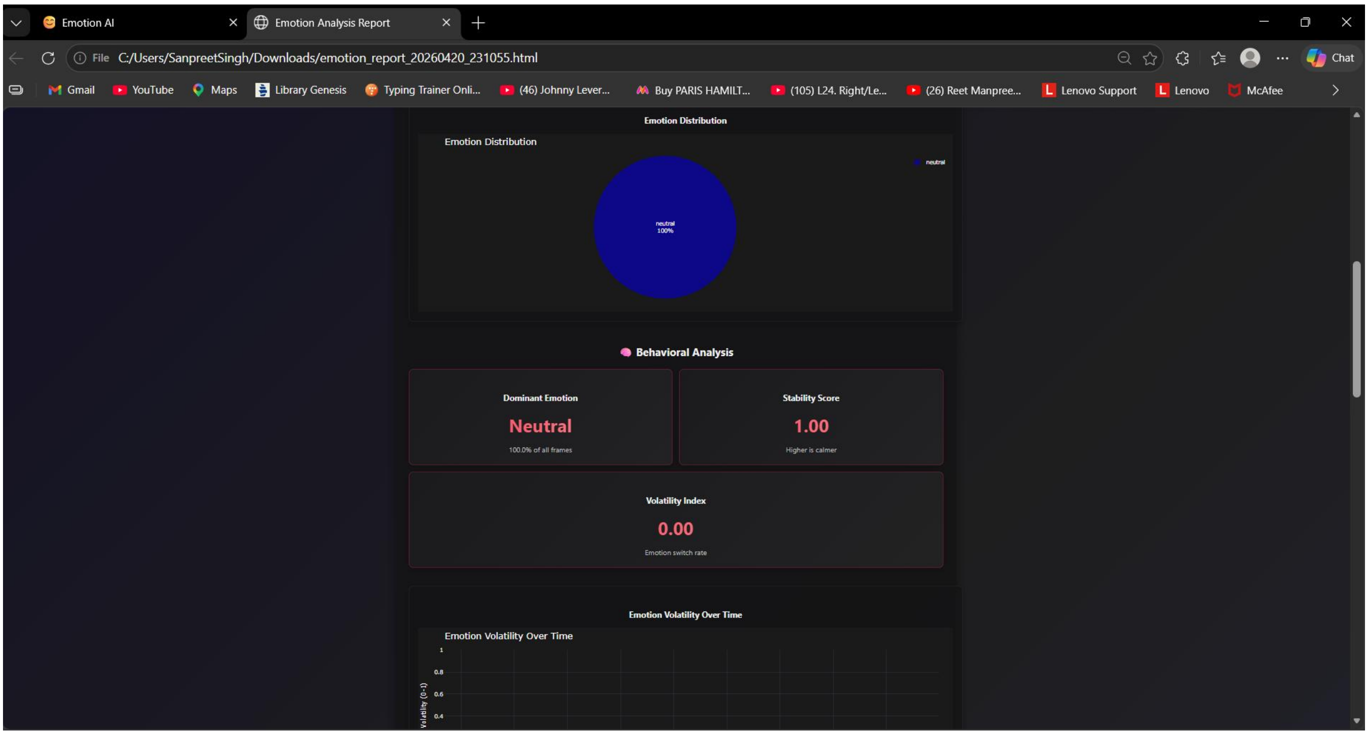
Allows users to generate and download comprehensive reports:

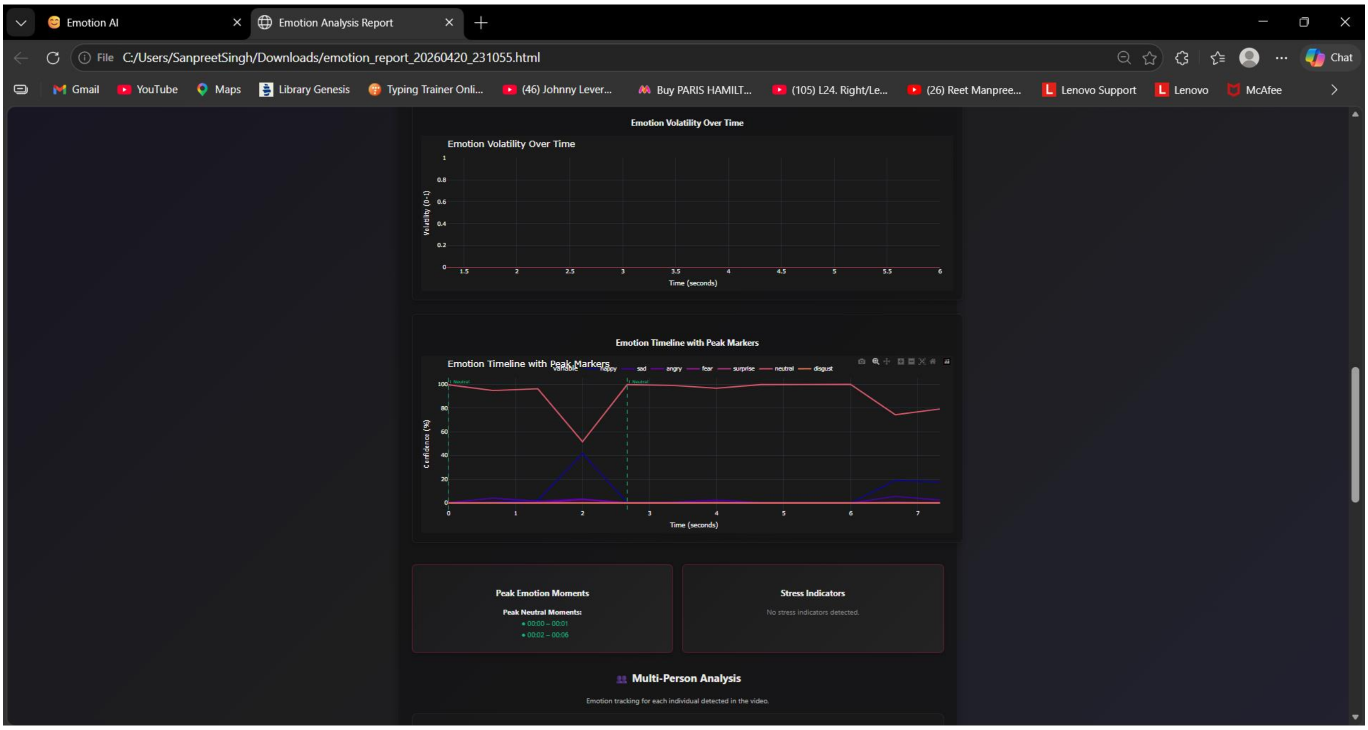
- HTML report generation with embedded charts and analysis
- Includes emotion timeline, distribution, and predictions
- Behavioral and multi-person analysis summaries
- Structured layout for easy interpretation and sharing

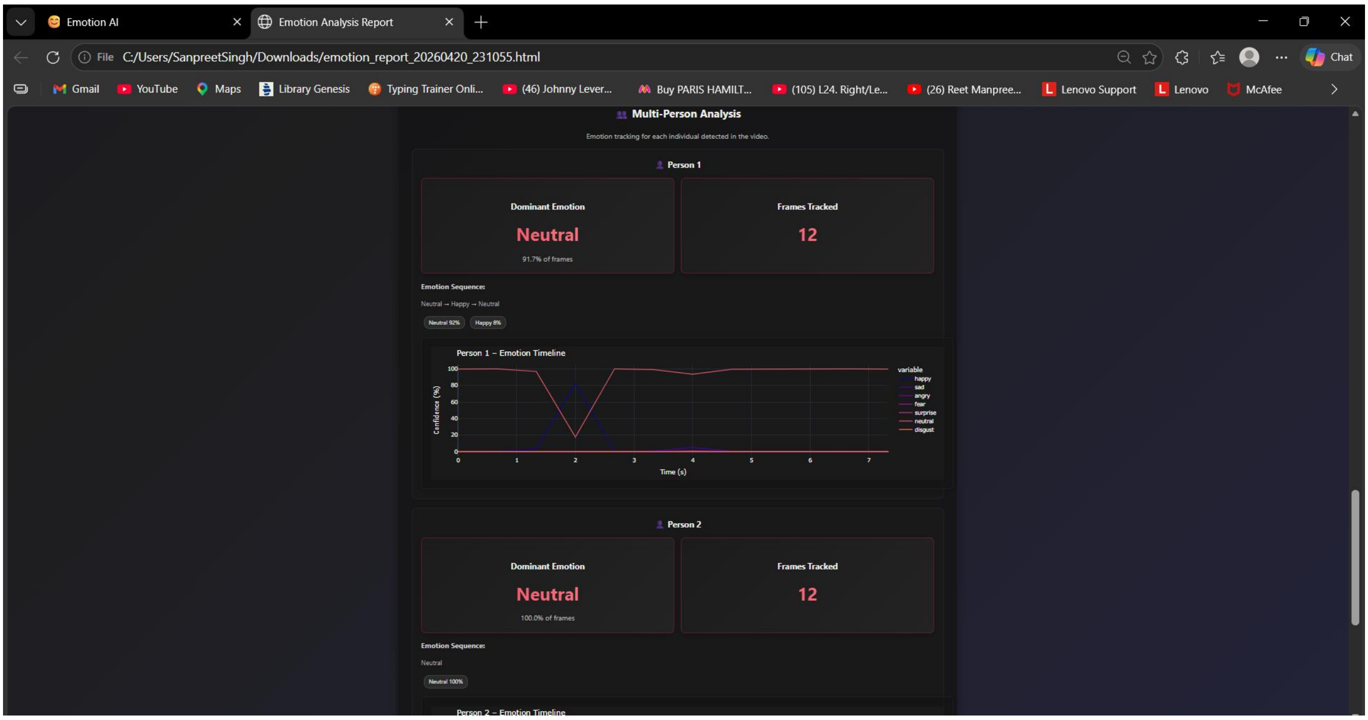

0
1
2
3
4
5
6
7
Time (s)
Download Detailed Report
High Accuracy
Advanced AI algorithms ensure precise emotion detection in real-time.
Fast Processing
Optimized for speed with minimal latency for smooth interaction.
Privacy Focused
All processing happens locally. No data is sent to external servers.

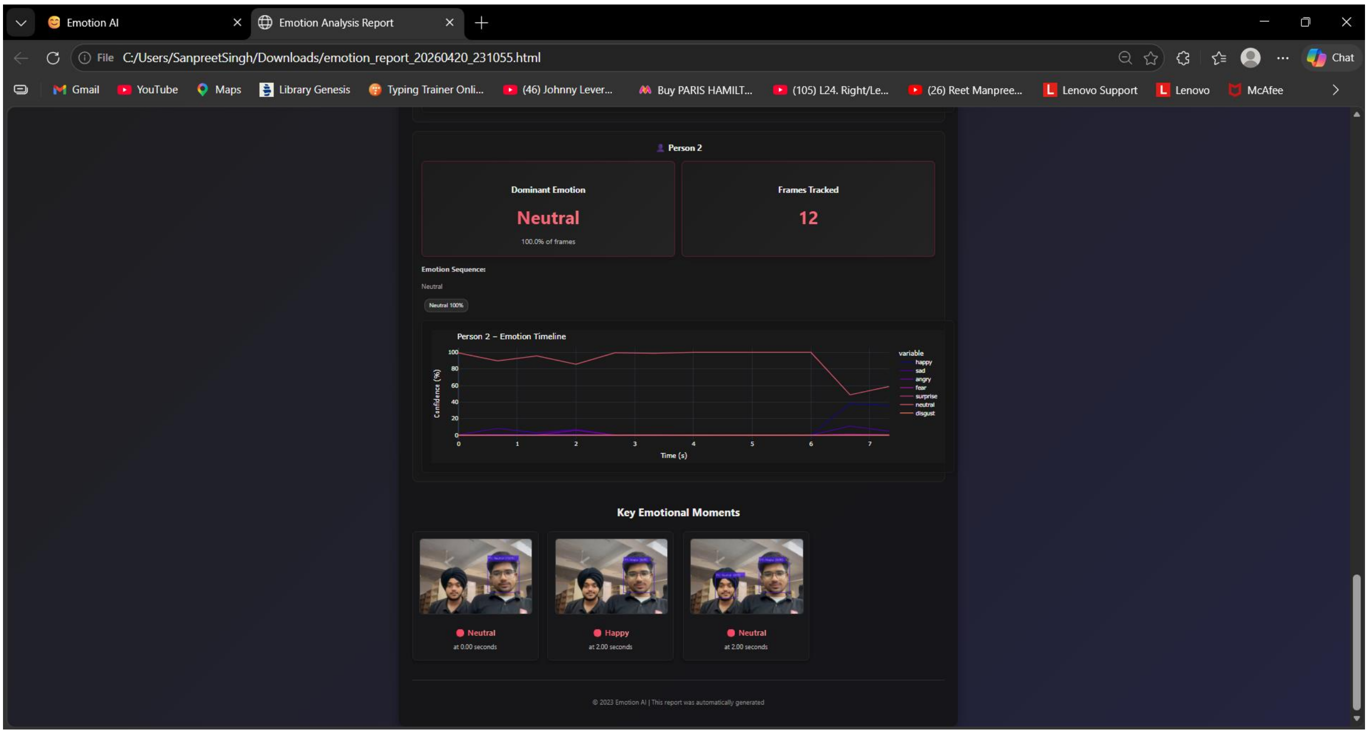

Chat
(105) L24. Right/Le...
(26)
Downloads
emotion_report_20260420_230537.html
Open file
See more
neutral
disgust
6
7


Emotion AI
Emotion Analysis Report
File C:/Users/SanpreetSingh/Downloads/emotion_report_20260420_231055.html
Video Emotion Analysis Report
Analysis generated on 2026-04-20 23:10:54
Duration
7.6s
Frames Analyzed
12
Dominant Emotion
Neutral
Emotional Analysis
Emotion Timeline
Emotion Confidence Timeline
happy
sad
angry
fear
surprise
neutral
disgust
Confidence (%)
Time (seconds)


Emotion AI
Emotion Analysis Report
File C:/Users/SanpreetSingh/Downloads/emotion_report_20260420_231055.html
Emotion Distribution
Emotion Distribution
neutral
neutral
100%
Behavioral Analysis
Dominant Emotion
Neutral
100.0% of all frames
Stability Score
1.00
Higher is calmer
Volatility Index
0.00
Emotion switch rate
Emotion Volatility Over Time
Emotion Volatility Over Time


Emotion AI
Emotion Analysis Report
C:/Users/SanpreetSingh/Downloads/emotion_report_20260420_231055.html
Emotion Volatility Over Time
Emotion Volatility Over Time
Volatility (0-1)
Time (seconds)
Emotion Timeline with Peak Markers
Emotion Timeline with Peak Markers
Confidence (%)
Time (seconds)
Peak Emotion Moments
Peak Neutral Moments:
00:00 – 00:01
00:02 – 00:06
Stress Indicators
No stress indicators detected.
Multi-Person Analysis
Emotion tracking for each individual detected in the video.


Emotion AI
Emotion Analysis Report
C:/Users/SanpreetSingh/Downloads/emotion_report_20260420_231055.html
Multi-Person Analysis
Emotion tracking for each individual detected in the video.
Person 1
Dominant Emotion
Neutral
91.7% of frames
Frames Tracked
12
Emotion Sequence:
Neutral → Happy → Neutral
Neutral 92%
Happy 8%
Person 1 – Emotion Timeline
Confidence (%)
Time (s)
variable
happy
sad
angry
fear
surprise
neutral
disgust
Person 2
Dominant Emotion
Neutral
100.0% of frames
Frames Tracked
12
Emotion Sequence:
Neutral
Neutral 100%
Person 2 – Emotion Timeline



5. CODE-IMPLEMENTATION AND DATABASE CONNECTIONS

The EmoGalaxy system is designed using a modular and scalable Python-based architecture, leveraging computer vision, machine learning, and audio processing technologies to deliver real-time multimodal emotion recognition through an interactive user interface. The implementation is divided into frontend interface components and backend processing modules.

Frontend Implementation (Streamlit + Custom Styling)

- The graphical user interface is developed using **Streamlit**, enabling rapid creation of interactive web applications.
- Custom styling is applied to enhance the visual appearance, providing a clean, responsive, and user-friendly interface.
- The application is structured into three main functional tabs:
  - **Live Camera**: Real-time emotion detection using webcam and microphone
  - **Upload Image**: Emotion detection for static images
  - **Video Analysis**: Advanced analysis with charts, insights, and report generation
- Interactive elements such as status indicators, expandable sections, and dynamic charts improve usability.
- Real-time updates and smooth transitions provide a seamless user experience.

Backend Implementation (Python-Based Processing)

- The core logic of the system is implemented in Python using a modular architecture.
- Emotion detection is performed through an integrated processing pipeline consisting of:
  - **OpenCV** for face detection and frame processing
  - **DeepFace** for facial emotion classification
  - Frame-by-frame analysis with temporal smoothing for stable predictions
- Video and image inputs are processed using optimized pipelines with adjustable sampling rates.
- Multi-person tracking is implemented using centroid-based matching to maintain identity across frames.
- Emotion data is continuously processed and stored temporarily for analysis and visualization.

Voice Emotion Detection Module

- The system incorporates a voice-based emotion detection module using:
  - **librosa** for audio feature extraction (MFCC, pitch, energy, spectral features)
  - **sounddevice[6]** for real-time audio recording
- Extracted features are analyzed to classify emotions from speech signals.
- This module enhances emotion detection by capturing non-visual emotional cues.

Multimodal Fusion Module

- Combines facial and voice emotion predictions using a weighted fusion mechanism:
  - Face Emotion contributes higher weight
  - Voice Emotion complements the prediction
- Generates a final, more accurate and context-aware emotion output.
- Handles missing inputs gracefully when one modality is unavailable.

Behavioral Analysis Module

- Analyzes emotion patterns over time to generate meaningful insights:
  - Volatility Index (frequency of emotion change)
  - Stability Score (emotional consistency)
  - Detection of stress indicators
  - Identification of peak emotional moments
- Provides analytical charts and summaries for better understanding of emotional behavior.

Emotion Prediction Module

- Implements a Markov Chain-based model to predict future emotional states.
- Builds a transition probability matrix from observed emotion sequences.
- Provides:
  - Next emotion prediction
  - Confidence scores
  - Multi-step emotion forecasting
- Visualizes predictions using interactive charts.

Report Generation

- Generates detailed **HTML reports** containing:
  - Emotion distribution charts
  - Timeline graphs
  - Behavioral analysis insights
  - Multi-person tracking data
  - Emotion prediction results
- Reports are downloadable and structured for easy interpretat

6. Conclusion

The EmoGalaxy project successfully demonstrates the integration of artificial intelligence, computer vision, and audio processing technologies to develop an advanced multimodal emotion detection system. The system is capable of analyzing emotions in real-time and from pre-recorded inputs using facial expressions, voice signals, and behavioral patterns.

By incorporating features such as multimodal emotion fusion, multi-person tracking, behavioral analysis, and emotion prediction, EmoGalaxy goes beyond traditional emotion detection systems and provides a more comprehensive and context-aware understanding of human emotions. The use of powerful libraries such as DeepFace, OpenCV, librosa, and Plotly, combined with Streamlit's interactive capabilities, results in a system that is both efficient and user-friendly.

The modular architecture ensures scalability and ease of maintenance, allowing for future enhancements such as cloud integration, personalized analytics, and mobile deployment. This project highlights the growing importance and potential of emotion-aware intelligent systems in fields such as mental health monitoring, smart surveillance, human-computer interaction, and user experience research.

Overall, EmoGalaxy represents a significant step toward building intelligent systems that can understand and respond to human emotions in a more accurate and meaningful way.

7. Future Scope

- **Emotion History Logging:** Store sessions in a database to build emotion trends for individuals.
- **Multi-modal Emotion Detection:** Integrate speech tone or text sentiment with facial cues.
- **Mobile Application Development:** Extend functionality to Android/iOS using Streamlit or Flutter wrappers.
- **User Profiles & Personalization:** Tailor analysis reports or recommend actions based on user history.
- **Live Streaming Integration**: Support real-time emotion tracking in Zoom/Meet via browser plugin.
- **Hybrid Model Training:** Fine-tune DeepFace on region-specific datasets for better accuracy.
- **Cloud Deployment:** Host on AWS, Azure, or Hugging Face Spaces for scale and accessibility.

8 REFERENCES

1. **DeepFace Library Documentation**

   [DeepFace Documentation]

   Official documentation for accessing photo used in the project.

2. **OpenCV Official Documentation [2]**

   [OpenCV Official Documentation]

   Provides comprehensive resources and best practices for using sc in computer vision, including face detection and image processing.

3. **Scikit-learn: Machine Learning in Python**

   [Scikit-Learn Official Documentation]

   Library used for implementing machine learning algorithms like TF-IDF and cosine similarity.

4. **Cosine Similarity Explanation and Usage**

   [Cosine Similarity Official Documentation]

   Wikipedia article explaining the concept and application of cosine similarity in text analysis.

5. **TensorFlow / Keras**

   [TensorFlow Documentation]

   Deep learning frameworks used for building and training neural network models.

6. **SoundDevice**

   [SoundDevice Documentation]

   A Python library used for recording and playing audio signals.